

APPROVED

BEng (Hons) Mechanical Engineering Programme
School of Computing, Engineering and Technology

Version	3.0
Course Type	Undergraduate
Award Type	Bachelor of Engineering with Honours - BEng (Hons)
Awards	
No Course Award Assigned	
Code	0469
SCQF Level	Level 10
QAA Subject Benchmark	Engineering (2023)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2026 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100190	mechanical engineering	mechanical engineering	100%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

The Mechanical Engineering courses (including the Mechanical and Electrical Engineering, and Mechanical and Offshore Engineering routes) cover an exciting blend of theory, practice and professional skills that will enable the student to develop a broad understanding of mechanical and electrical engineering topics and an in-depth knowledge of modern engineering technologies. The course will provide students with the range of technical skills and knowledge needed to address real-world challenges. In addition to providing the student with the skills that reflect the professional standards demanded by industry, the course offers a variety of specialised topics such as renewable and conventional energy technologies, Failure Analysis, and Control and Instrumentation.

The course involves a broad range of teaching techniques. These include lectures covering key theories, tutorials where students apply theory to solve practical problems, and laboratory

What the Course Involves	sessions involving the investigation and analysis of engineering principles and systems. Students may also undertake an industrial placement. Hands-on learning is a key feature of the course, with extensive use of workshops, CAD and simulation tools, and practical activities. Teaching techniques are selected based on the suitability to the particular subject. It may consist of the staff supporting the students to engage in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities. Generally, teaching is used to discuss the underlying theories, support the peer-learning, clarify the material and experiment with the techniques and skills discussed. At each stage of the course, project-based learning provides the foundation for applying theoretical knowledge to real engineering challenges. The ability to work as an effective member of a project team is an essential skill for anyone working in industry, and group working is a major feature of the course. In each semester, students are typically expected to attend three hours per week per 15 credit module in lectures/tutorials/laboratories. Delivery follows the University Academic Calendar , with timetabled sessions during the teaching weeks. Students are expected to continue their learning through independent study activities outside timetabled sessions. This course offers the opportunity for students to further tailor their course to their interests through the choice of elective modules. The elective modules enable students to gain an appreciation for the wider context within which Engineering operates and diversify their skill set. Students can undertake an optional placement. Placements are an important learning tool. The interaction with the staff of the company, the required tasks and the general environment provides an experience difficult to achieve by other means. Students are supported on placements by a course tutor from the staff of the School and an industrial tutor from the company. The School tutor supports and monitors the assessment process.
UCAS Code	Code H301 H304
Course Collaborations	
No collaborations listed	
Introduction to the Course	The BEng (Hons) Mechanical Engineering courses are designed to produce future-proof professional engineers who are capable of tackling existing and future engineering challenges and creating solutions to real-world problems. Mechanical Engineering underpins many sectors, including energy, transport, manufacturing, and healthcare. This course will provide students with comprehensive knowledge and skills in mechanical design, thermofluids, materials, dynamics, and control. It also emphasises sustainability as a core principle, recognising the pivotal role that engineers play in tackling global challenges. The curriculum combines technical skills with a deep understanding of environmental and social responsibility aligned with the United Nations Sustainable Development Goals. A key feature of this course is the integration of practical laboratory work and project-based learning in all stages, which enables students to develop industry-relevant skills, creativity and other key qualities of professional engineers through team and individual work. In this course, the student will undertake a variety of modules to learn underpinning engineering principles, science and mathematics. The course includes comprehensive study of classical and modern mechanical systems, fluid and thermal sciences, materials and manufacturing processes, and the design and operation of machines and mechanisms. Through interdisciplinary, practical and project-based learning, students will develop a wide range of technical and transferable skills to meet the challenges and rapid changes in the industry. This course enables students to progressively engage in the development of the engineering management, environmental, innovation and sustainability skills demanded by the industry and to become a life-time learner. The final stage of the course enables students to embrace advanced technology in renewable energy engineering and specialise in the area of their interests through a final-year individual project. Building on the School's research strengths in mechanical systems, manufacturing, and energy engineering, and supported by strong links with local, national, and international industries, this course offers specialised topics and industrial engagement opportunities to broaden and deepen students' engineering perspectives.

	This course provides students with the industry-relevant knowledge, understanding, skills, attributes, and vision required by the engineering industry and professional bodies. Completing this course gives students the ability to work effectively across a wide range of industries, including manufacturing, energy, transport, and offshore engineering.
Educational Aims of the Course	The course aims to develop adaptable, forward-thinking engineers equipped with the knowledge, skills, and creativity to tackle complex challenges and drive innovation in the global renewable energy sector. Stage 1 (CertHE Engineering) Stage 1 aims to provide students with a solid grounding in the scientific, mathematical, and engineering principles needed by engineers. This stage introduces key concepts in sustainability and inclusive design, while developing technical competence and problem-solving skills. It prepares learners for progression to more advanced study and practical application within the field of engineering. Stage 2 (DipHE Mechanical Engineering) Stage 2 aims to deepen students' understanding of engineering science and mathematics and to introduce key Mechanical Engineering technologies. This stage builds on foundational knowledge to develop competence in evaluating, comparing, and optimising renewable energy solutions. Learners will also strengthen their awareness of sustainability, and the broader environmental and societal impacts of energy choices. Stage 3 (BEng Mechanical Engineering; BEng Mechanical and Electrical Engineering; BEng Mechanical and Offshore Engineering) Stage 3 aims to develop students' ability to apply advanced engineering principles and analytical techniques to the design, integration, and evaluation of Mechanical, Electrical, and Offshore systems. This stage emphasises independent learning, critical thinking, and interdisciplinary problem-solving, enabling students to tackle real-world engineering challenges. Learners will also enhance their understanding of professional practice, sustainability, and the ethical responsibilities of engineers. Stage 4 (BEng (Hons) Mechanical Engineering; BEng (Hons) Mechanical and Electrical Engineering; BEng (Hons) Mechanical and Offshore Engineering) Stage 4 allows students to demonstrate independent, critical, and creative thinking in the design, evaluation, and implementation of Mechanical, Electrical, and Offshore systems. This stage focuses on applying advanced engineering knowledge to complex, real-world challenges. Learners will undertake substantial individual projects, refine professional and research skills, and prepare for graduate-level employment or further academic study.
Awarding Institution	Robert Gordon University
Course Duration	Full-time: BEng (Honours) - 4 academic years (8 semesters). Part-time: In general part-time study takes twice as long as full-time. BEng (Honours) - 8 academic years (16 semesters).
	The BEng (Hons) Mechanical Engineering courses (including the Mechanical and Electrical Engineering, and Mechanical and Offshore Engineering routes) have a strong practical focus on the design, implementation, and optimisation of sustainable energy systems and the technologies that support them. A central theme of the courses is the integration of analytical, computational, and experimental approaches to solve complex engineering problems. This approach fosters a culture of innovation and problem-solving, empowering graduates to confidently tackle real-world energy challenges with the technical and creative expertise demanded by the global energy sector. The course takes a pragmatic approach to developing the student's readiness for modern life and future employment, supporting the student in expanding the skills essential for responding to future technological and societal changes. Through the utilisation of individual and group work and project teams, students are guided in cultivating skills that enhance the

Distinctive Features of the Course	students' employability, such as: self-management abilities, such as focus, integrity, adaptability, and initiative; social intelligence by developing skills in communication, collaboration, and leadership; and innovation by fostering qualities such as curiosity, creativity, sense-making, and critical thinking. The course is taught by a team of full-time staff who have extensive industrial, commercial and research experience in engineering. The course enables the students to benefit from their experience and involvement in the leading edge of the development and application of renewable energy technologies. This course provides the student with the chance to customise their learning experience by selecting elective pathways focused on specific engineering themes. This allows students to understand engineering as a multidisciplinary subject and build their community of practice across the School. The course also provides the students with the option to undertake a study abroad placement between Stage 2 and Stage 3 of study. The School partners with a number of European institutions and has recently participated in exchanges with Belgium, Finland, Norway and Sweden.
Placement/Professional Experience	Students may elect to undertake a short summer placement in a summer break and/or a year-long placement between stages. The placement provides an opportunity to sample working life and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who the student can contact about any academic or non-academic problems during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short summer placement between Stages 1 and 2, or Stages 2 and 3, or Stages 3 and 4. This normally comprises at least 4 weeks of continuous, full-time supervised work experience. The assessment for a short summer placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short summer placement will normally receive at least 1 video call with their Academic Tutor. In addition, or as an alternative to a short summer placement, students may undertake a year-long placement either between Stage 2 and 3, or between Stage 3 and 4. The year-long industrial placement normally comprises of 30 - 48 weeks of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. Students may elect to undertake a Study Abroad placement between stages. This is typically a year-long option. The Study Abroad placement is to be undertaken with one of the School's pre-approved partner institutions. Whilst the contents of the Study Abroad placement vary on an annual basis and depending on the host institution, students are expected to complete their Learning Agreement pre-departure, with a focus on choosing modules which complement their course of study at RGU.

	All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 120 credits achievable from taught modules at each stage. Successful completion is not required for progression to the following stage. The credits achieved for placement may not be used to compensate for any credits from the taught modules. For students on the MEng fast track route, two summer placements are compulsory, undertaken in the summers following stages 3 and 4. These assessed placements form an integral, credit-bearing part of the course.
Opportunities for Further Study	An Honours degree achieved with at least an upper second class (Class 2.1) Classification will normally provide access to postgraduate and doctoral programmes in UK higher education institutions.
External and Internal Indicators of Quality and Standards	The Quality Assurance Agency for Higher Education (QAA) has published the Engineering Benchmark Statement, which outlines the standards expected for the award of Honours and Master's Degrees in Engineering. The QAA Benchmark Statements are used during the design of modules, courses, and the Undergraduate and Integrated Master's Engineering Programmes. The structure and content of these courses have been constructed following the QAA's Engineering Benchmarks and principles of course design. The content of the courses has been selected to provide a balanced coverage of the broad categories of abilities and skills identified in the QAA Engineering Benchmarks: Engineering-related cognitive abilities, Engineering-related practical abilities, and Additional transferable and professional skills. These have been formulated as learning outcomes within the Scottish Credit and Qualifications Framework (SCQF). The course design has also been strongly influenced by key engineering concepts: the application of scientific principles, the integration of design and analysis, the use of modern computational tools, and the balance between theory and practice in solving real-world problems. SCQF credit values have been used as a guide during the design of modules, while the structure and content of the courses have also been aligned with the Institution of Mechanical Engineers (IMechE) and the Institution of Engineering and Technology (IET) accreditation requirements for professional recognition. Embedded in the core structure are modules and topics that provide students with a broad foundation in engineering science, design and methodology, tools and techniques, project management, and professional practice. Standards are primarily established through QAA Benchmarks and the requirements imposed by the IMechE and IET in order to gain professional course accreditation. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's degree programmes, and the scope of the subject discipline, as defined by QAA, is fully reflected in the current provision within the School's engineering portfolio.

	A number of internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholder views on standards are routinely obtained through the Staff–Student Liaison Committee, the Course Management Teams, Student Voice questionnaires, the Industrial Advisory Board, industrial visits to placement students, and staff involvement in professional body activities and external accreditation events. Best practice in teaching and learning within the University is monitored through the University's quality enhancement and assurance processes, which include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the University's quality assurance committee structures, such as the Student Experience Sub-Committee and the Academic Regulations Sub-Committee. Finally, it should be noted that the academic staff within the School of Computing, Engineering and Technology have a continuing interest in pedagogical research and a commitment to the scholarship of engineering education. Staff are active in organising and participating in national and international conferences on engineering education and have contributed to research projects funded by bodies such as the Higher Education Academy, the Royal Academy of Engineering, and the Scotland's Tertiary Enhancement Programme (STEP), as well as through the University's own Teaching and Learning Conferences.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	SQA Higher: BBCC to include Maths and either Engineering Science, Physics, Chemistry or Technological Studies. English at National 5 grade C or above is required if not held at Higher. GCE A Level: CCC to include Mathematics and one from Physics, Chemistry, Design Technology, Electronics, or Engineering. GCSE English at grade 5/C or above is required if not held at A Level. Irish Leaving Certificate Higher: H3H3H3H3 to include Maths and either Physics, Chemistry or Technological Studies. English at grade O3 or above is required if not held at Higher. IB Diploma: 26 to include Higher Level Maths and Physics or Chemistry at grade 5. English is required at a minimum of Standard Level grade 4. Minimum Entry requirements for widening access applicants: SQA Higher: CCCC to include Maths and either Engineering Science, Physics, Chemistry or Technological Studies. English at National 5 C is required if not held at Higher.

Alternative Qualifications for Entry	We welcome applications from those with equivalent qualifications to those stated, including SVQs/NVQs, access programmes and overseas qualifications. Applications from students with non-standard qualifications or work experience will be considered on an individual basis. Appropriate HNC or HND may allow entry to Stage 2 or Stage 3. These are considered on an individual basis.
Interview Procedures	We do not normally conduct interviews as part of our admissions process. However, we invite all applicants to a Discovery Day so that we can show them around the School, answer any questions they might have about the course and discuss their future career opportunities. This includes an informal one-to one discussion with a member of academic staff. Interviews may be required on an individual basis for non-standard entry requirements.
Course-Specific Regulations	Honours Classification is based solely on the student's performance in Stage 4 of the course. The credits achieved for placement or study abroad modules are in addition to the credits achievable from taught modules and may not be used to compensate for any credits from the taught modules. Electives will run at the discretion of the School. Condonement <i>The Assessment Board has discretion to condone a partial failure in a module in accordance with Regulation A4 and providing the Engineering Council requirements, as detailed within their Compensation and Assessment Policy, are met:</i> • <i>The partial failure is at grade E.</i> • <i>All aligned AHEP learning outcomes are met in other successfully completed modules.</i> • <i>The total volume of condoned credit does not exceed 30 credits across all stages, excluding Stage 1.</i> Modules Classed as Central to the Objectives of the Course The requirements of the Engineering Council are that, in order to achieve an accredited Honours award, all major project modules must be passed. The capstone project module has been identified as being central to the objectives of the courses in the programme and, therefore, failure may not be condoned [refer to Regulation A4, paragraph 11.1]. Transfer to MEng Stage 3

	There is the opportunity for students to transfer from the BEng (Hons) to the MEng course. In order to be invited to transfer to Stage 3 of the BEng Mechanical Engineering course, students should achieve at least a B profile overall at the end of Stage 2. A B profile should have at least 50% of modules at Grade B or better, at least 75% of the modules at Grade C or better, and normally 100% of modules at Grade D or better. Transfer to MEng Stage 4 In order to be invited to transfer to Stage 4 of the BEng Mechanical Engineering course, students should achieve at least a B profile overall at the end of Stage 3. A B profile should have at least 50% of modules at Grade B or better, at least 75% of the modules at Grade C or better, and normally 100% of modules at Grade D or better.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK.

	RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
	The BEng (Hons) Mechanical Engineering courses (including the Mechanical and Electrical Engineering, and Mechanical and Offshore Engineering pathways) have been specifically designed to create modern engineers who can draw on their knowledge and skills to deliver innovative, sustainable, and efficient solutions to real-world problems. Within the courses there is adherence to three key principles: • To encourage innovation through investigation and understanding of mechanical, electrical, and offshore engineering systems and technologies. • To enhance the process of engineering design and analysis, integrating theoretical, computational, and experimental approaches. • To demonstrate the practical application of skills and knowledge in engineering design, manufacturing, and problem-solving through hands-on laboratory and project work. A feature of these courses is the optional industrial or study-abroad placement. This provides students with the opportunity to gain advanced professional skills through industrial experience with an engineering company or to study at a partner university abroad, gaining valuable insight into international engineering practice. The course modules specifically address these principles through the following main themes in engineering. Design and Innovation Design is central to all branches of engineering. The courses develop students' ability to apply design methods to mechanical, electrical, and offshore systems—from concept generation and analysis to optimisation and testing. Students use industry-standard tools such as CAD, CAE, and simulation software to model and evaluate designs. They learn to incorporate sustainability, manufacturability, and safety into their design decisions, preparing them to deliver creative and practical engineering solutions. Analysis and Simulation A key feature of the courses is the strong emphasis on analytical and computational methods. Students learn to apply mathematics, mechanics, thermodynamics, and control theory to engineering systems, supported by simulation and numerical analysis using software such as MATLAB, ANSYS, and SolidWorks. This ensures that graduates are confident in interpreting complex data, validating results, and applying modern engineering tools to solve problems efficiently. Sustainability and Emerging Technologies Sustainable engineering is a core theme throughout the courses. Students learn about the environmental, social, and economic implications of engineering activities, developing the ability to design and evaluate technologies that support decarbonisation and resource efficiency. Topics such as renewable energy, automation, hydrogen and electrification, and offshore energy systems are integrated across modules to reflect the changing priorities of modern engineering practice.

Course-Specific Learning Experience

Professional and Interpersonal Skills

The courses place strong emphasis on the development of professional, management, and interpersonal skills. Through individual and group projects, students build confidence in project planning, communication, and leadership. Modules embed topics such as engineering ethics, risk assessment, and health and safety. The courses culminate with a major project, either individual or group, through which students demonstrate their ability to design, implement, and evaluate a substantial engineering solution, documenting and presenting their work to professional standards.

External Competitions and Challenges

As part of the learning experience, students are encouraged to participate in external competitions and professional challenges. These may include:

- **Formula Student** , where students design, build, and race a single-seat racing car.
- **The IMechE Design Challenge and MATE ROV** , which promotes practical problem-solving and teamwork.
- **Renewable energy or robotics competitions** , run by national or international engineering organisations.

Participation in such events helps build students' confidence, creativity, and teamwork skills in a competitive yet supportive environment.

Industry Links and Visits

The courses benefit from strong partnerships with local and national industry. Regular **guest lectures, site visits, and industrial projects** expose students to real engineering environments and professional practices. Industrial partners also contribute to module delivery, offer placement opportunities, and provide feedback through the School's Industrial Advisory Board, ensuring that course content remains aligned with current industry needs.

Technical Facilities

Many of the practical aspects of the courses take place in the School laboratories. The School houses a comprehensive suite of facilities, including:

- Mechanical testing laboratories for materials and structural analysis.
- Thermofluids and energy systems laboratories with modern wind tunnels, flow rigs, and heat-exchange test stands.
- Electrical and control systems laboratories equipped with programmable logic controllers (PLCs), sensors, and automation rigs.
- CAD and simulation suites with access to leading software packages such as **SolidWorks** , **ANSYS** , **MATLAB** , and **Simulink** .
- Workshops for prototype manufacture, 3D printing, and fabrication.

Laboratories are continually updated on a rolling programme to ensure that facilities remain state-of-the-art and aligned with industrial standards. Access to the labs is by electronic key fob during University opening hours, and wireless connectivity allows students to link directly to the University network and software resources.

Software Provision and Digital Learning

Software provision is comprehensive, covering design, simulation, and analysis tools. Standard installed packages include **SolidWorks** , **ANSYS** , **MATLAB/Simulink** , and the **Microsoft Office 365** suite. Much of the software used on the courses is available under academic or educational licences, allowing students to continue their work off-campus with the same setup as provided in the labs.

This combination of **strong theoretical foundations** , **hands-on laboratory experience** , **industrial engagement** , and **emphasis on design, analysis, and sustainability** ensures that graduates are well prepared to contribute immediately and effectively to the engineering profession.

The Learning Process: Teaching Strategy and Methods

The course curriculum is designed as a coherent whole with an emphasis on the integration of **engineering theory, practical skills, and real-world applications** . It enables the student to acquire and develop the knowledge, understanding, and professional competences necessary for successful employment as an engineer, and provides a disciplined framework for problem definition, specification, design, analysis, implementation, and evaluation.

Such knowledge includes an understanding of the range of possible options for these tasks, appropriate engineering and digital technologies, and the practical and transferable skills required to work collaboratively and ethically in a multidisciplinary environment. The curriculum also provides the underpinning knowledge necessary to sustain lifelong learning, professional adaptability, and continued development in an era of rapid technological change.

These skills are developed progressively across the **Scottish Credit and Qualifications Framework (SCQF) Levels 7 to 10** , corresponding to the BEng (Hons) awards.

A general feature of the courses is the emphasis on a unified, integrated view of engineering. This is reflected in the following design principles:

- Presenting a coherent underpinning of theoretical and practical knowledge that develops progressively throughout the programme, enabling graduates to adapt to future developments in engineering.
- Reflecting the rapid evolution of engineering technologies and ensuring coverage of emerging topics such as renewable energy, automation, and sustainability.
- Integrating delivery across disciplinary boundaries to highlight the interconnectedness of mechanical, electrical, and offshore systems.
- Embedding varied modes of assessment to balance formative and summative evaluation, individual and group work, and practical and theoretical tasks.
- Encouraging students to combine and synthesise learning from different modules through integrated and synoptic assessments, demonstrating a holistic understanding of engineering systems.

The professionally orientated nature of the courses requires the student to develop both **general and discipline-specific skills** , achieved through a combination of teaching and learning methods including lectures, tutorials, design studios, laboratories, workshops, seminars, industrial placements, integrative project work, student-centred learning, and independent study.

Teaching and Learning Approaches

The main goal of the teaching and learning strategy is to enable students to become **critical, reflective, and independent learners** who are active participants in the learning process. Students are encouraged to take increasing responsibility for their learning as they progress through the course, developing the motivation and autonomy needed to evaluate their own effectiveness as engineers and to engage in lifelong learning.

This process is underpinned by the constructive alignment of learning outcomes, syllabus content, and assessment strategy, delivered through a diverse range of teaching methods. These include conventional instructional approaches, collaborative group and team-based learning, and experiential and enquiry-based learning such as design projects, problem-based learning (PBL), and laboratory investigation.

Teaching and Learning Strategy

Extensive use of **technology-enhanced learning** is made throughout the programme. Examples include simulation-based tutorials, digital laboratories, and virtual learning resources available via the University's Virtual Learning Environment (Moodle). Assessment and feedback are supported through online tools and automated feedback systems, enhancing engagement and promoting self-reflection. Integrated teaching sessions often combine short periods of theoretical instruction followed by immediate practical or simulation-based exercises, reinforcing learning through direct application.

Modes of Delivery

Lectures

Lectures are used for the dissemination of key theoretical concepts, demonstration of engineering methods, and the discussion of principles and their applications. They are supported by visual aids, digital projection, and simulation demonstrations. Opportunities for student participation and discussion are actively encouraged.

Tutorials

Tutorials provide an environment for students to practise analytical and design techniques under academic guidance. These may take the form of problem-solving sessions, design clinics, or group-based exercises where students work collaboratively to apply engineering theory to practical challenges. Tutorials also provide opportunities for formative feedback and clarification of lecture material.

Laboratories and Workshops

Practical experience is essential to engineering education. Laboratory sessions allow students to develop hands-on skills in measurement, experimentation, analysis, and interpretation of data. These activities take place in mechanical testing, thermofluids, materials, and control laboratories under staff supervision. Workshops and CAD suites are also used for design and prototyping activities. The "mini-lecture and lab" model is increasingly employed, where short instructional sessions are interspersed with practical exercises to consolidate learning.

Industrial Experience

The course includes optional placements that provide invaluable real-world experience. Students competitively apply for placements by submitting CVs and attending interviews. While on placement, they undertake professional tasks under the supervision of an industrial mentor and are supported by an academic tutor from the School. These experiences are further enhanced through a series of preparatory seminars and debriefing sessions delivered by academic staff, careers advisers, and industry professionals.

Group Projects

Group projects simulate professional engineering practice and are undertaken under staff supervision. Students work in small, multidisciplinary teams to complete design, analysis, or manufacturing challenges. These projects develop technical competence, teamwork, communication, leadership, and project management skills. Deliverables typically include design documentation, technical analysis, and a prototype or model, accompanied by professional reports and presentations.

Capstone / Major Project

	In the final stage of the BEng (Hons) courses, students complete a substantial individual or group project under academic supervision. Projects involve the design, modelling, analysis, or experimental investigation of an engineering system, culminating in a formal report and presentation. This major project replicates the professional responsibilities of a practising engineer and provides an opportunity to demonstrate integrated learning, innovation, and technical mastery. Student's Role in Learning Students are expected to be active participants in their learning journey. They are encouraged to extend their learning beyond formal instruction through wider reading, independent research, and practical enquiry. As they progress through the programme, students are expected to become increasingly self-directed and self-motivated, taking responsibility for their academic and professional development. Regular reflection on learning progress is promoted through tutorials and project supervision, helping students to identify areas for improvement and consolidate strengths. Students are expected to make full use of the wide range of facilities and resources available to them, including laboratories, workshops, software tools, and the University library and online databases.
	The Assessment of Student Work A variety of assessment methods are used in this course, as deemed appropriate to the subject material being assessed and the context in which the material is delivered. The University has developed an Assessment and Feedback Standard, which guides all assessment practices on the course. This ensures the student has opportunities to engage with assessment and feedback which extends and enhances their knowledge and skills; offers a balanced and manageable assessment load; and provides good quality, effective feedback. Assessments are developed so that they ensure module and course-wide learning outcomes have been met corresponding at the appropriate SCQF Level 7-10 for the course stage with respect to the characteristics of knowledge and understanding; practical skills; generic cognitive skills; communication, numeracy and IT skills, autonomy, accountability and working with others. It is important to note that all grades indicated to the student during the course of the academic session are provisional until the deliberations of the Assessment Board for the course. Provisional grades are presented to this Board, which consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each module undertaken by them. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal. The School Office prepares a schedule of coursework assessment and feedback deadlines. The assessment timetable is structured to ensure workload and deadline tensions are suitably managed. The student is informed how their assessment will be scheduled and the criteria made transparent in the context of coursework and grading structures. Computer Aided Assessment Computer-assisted assessment packages integrated into the virtual learning environment such as Moodle's question system are used in many modules, both for formative and summative assessment, to enhance student engagement and increase the speed of feedback. Peer Assessment Peer assessment is utilised in many modules throughout the course where group work is undertaken. This serves to develop working and interpersonal skills, as well as honing appraisal and critique skills. Peer assessment may be used in both summative and formative contexts.

Assessment	Unseen Exams In general unseen exams are used to assess the student's knowledge and understanding, and as the student progresses through the course unseen exams feature as a component of the assessment of taught modules. Assessed Laboratories Assessed laboratories are used to assess the student's knowledge, understanding and practical skills, and generally involve the student undertaking an unseen practical task such as developing a part of an information system, under open or closed book, supervised conditions with a fixed time constraint. Coursework For summative assessment the achievement of Learning Outcomes is assessed by coursework or a combination of exam and coursework. Coursework may take many forms, i.e. reports, essays, production of software, portfolio of work, presentations, design documentation and modelling activities. Projects/Dissertations Extended projects are used in later stages as a learning vehicle and assessment process that allow the student to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. These projects often require the practical application of a number of different techniques and technologies and provide the students with the opportunity to learn new investigative and evaluative skills. Projects, in particular group projects, provide an opportunity to learn new investigative and evaluative skills, and are also often used as the vehicle for gaining valuable employability skills. Project work is usually assessed by a combination of assessment of the completed project deliverable as an artefact, accompanying report or dissertation, and verbal presentation and demonstration documenting the project work undertaken by the student. Assessment Types Some modules use a combination of assessment types that integrate modes of assessment in order to balance the types of activity (e.g. individual and group activities) and purposes of assessment (e.g. formative and summative). Where there are a number of modules that cover allied subject material, integrated assessment may be used to facilitate the student to exhibit competence and achievement across a number of different themes. In all stages, synoptic assessments embedded into projects and other practical assignments are used to encourage the student to combine elements of their learning from different parts of the course and to show their accumulated knowledge and understanding of a set of topics or a subject area. A student's Honours Classification (First Class Honours, Upper Second Class Honours etc.) is based solely on performance in Stage 4 of the course. Equality and Diversity All assessment practices follow the University's Equality, Diversity, and Inclusion principles. Assessment materials and case studies aim to reflect a wide range of engineering contexts and ensure that fairness, accessibility, and inclusivity are maintained across all learning and evaluation activities.
	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>;

University Support for Teaching, Learning and Assessment	• an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	In addition to the University-wide support for Teaching and Learning, the School provides a number of means of supporting the student in teaching and learning. Members of staff publish course based teaching and learning materials on the University's Virtual Learning Environment, Moodle. The range of published materials spans from supplementary teaching materials such as additional lecture notes and tutorial guides to software development to general guidance for the student, such as codes of conduct in laboratories. Information on studying abroad can be accessed at http://www.rgu.ac.uk/future-students/study-abroad . Students are also supported by the SOCET support team who are able to advise students about a range of issues, both personal and academic, that may be of concern to the student or affecting their studies. The School runs a comprehensive induction programme during Welcome Week where the student will be introduced to the School, introduced to key members of staff. During the induction event, the student will be made familiar with the University's IT, learning environment and associated systems. Feedback is provided continuously to the student through informal contact with subject lecturers and tutors in the tutorial and laboratory settings. Feedback is provided on assessed practical work within 20 working days of submission of the work. Feedback on results can be provided by module staff, and facilities are made available for all students to view their marked examination scripts on request and under supervision. Grades are made available electronically for viewing via the student portal following ratification by the Assessment Board. The 'Student Handbook' provides details on the School, course, and working environment. It contains information regarding teaching and assessment, IT support, student support services, learning resources, including the library and health & safety. It details key contact names and email addresses. The handbook will be located on Moodle and will be organised into sections so that information is readily accessible.
	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process;

University Monitoring of Quality and Standards	• <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	This course fully complies with the above procedures. The course management team meets regularly to review any specific course issues, student feedback and module evaluation, External Examiners' reports and matters arising from Staff/Student Liaison meetings. Annual Course/Programme Appraisals are reviewed and approved by the Programme Management Team and School Academic Board which meet at least once per semester. Within the School of Computing, Engineering, & Technology the School's Teaching Committee regularly looks at module metrics such as student experience and student achievement to ensure that the models are continuing to deliver the required standard set by the School. The School's Teaching Committee, the Programme Management Team and School Academic Board also review all proposed changes to the structure and content of the course. The Undergraduate Engineering Programme is also informed by feedback from the School's Industrial Board.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level;

	• support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	Within the School of Computing, Engineering and Technology, there is a single Staff/Student Liaison Committee for all undergraduate degree courses. It normally consists of the Dean of School, Associate Dean for Academic Development and Student Experience and Course Leaders, with student representatives from each stage of each course. The Committee meets at least once per semester and feeds appropriate information to other departments and/or committees as required. Student representatives also take part in the School Academic Board where they can contribute to discussions about School policy. In addition, students are supported through the School's Team Around a Student (TAAS) approach, and there are regular scheduled meetings between Course Leaders and student representatives to discuss issues arising within specific classes and on individual modules. Student Voice Questionnaires are distributed to students on all modules during semester which allow them to raise issues and give feedback to staff.

Awards

Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Certificate of Higher Education - CertHE	Engineering	Aberdeen, United Kingdom	Exit Award		120

Stage 2

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Diploma of Higher Education - DipHE	Mechanical Engineering	Aberdeen, United Kingdom	Exit Award		240

Stage 3

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Engineering - BEng	Mechanical and Offshore Engineering	Aberdeen, United Kingdom	Exit Award		360
Bachelor of Engineering - BEng	Mechanical and Electrical Engineering	Aberdeen, United Kingdom	Exit Award		360
Bachelor of Engineering - BEng	Mechanical Engineering	Aberdeen, United Kingdom	Exit Award		360

Stage 4

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Engineering with Honours - BEng (Hons)	Mechanical Engineering	Aberdeen, United Kingdom	Target Award		480
Bachelor of Engineering with Honours - BEng (Hons)	Mechanical and Offshore Engineering	Aberdeen, United Kingdom	Target Award		480
Bachelor of Engineering with Honours - BEng (Hons)	Mechanical and Electrical Engineering	Aberdeen, United Kingdom	Target Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate understanding of basic concepts in physics, mathematics, and engineering	SC1	SCQF 7
LLO2	Show a basic awareness of the need to consider sustainability, equality, diversity, and inclusion in professional practice	SC1	SCQF 7, SCQF 7
LLO3	Undertakes practical workshop and laboratory activities in a safe manner.	SC2	SCQF 7
LLO4	Use standard laboratory equipment and software tools to carry out simple experiments and simulations.	SC2	SCQF 7
LLO5	Produce basic Engineering drawings	SC2	SCQF 7
LLO6	Analyse straightforward problems and select appropriate methods to reach solutions.	SC3	SCQF 7
LLO7	Communicate information clearly using a variety of formats.	SC4	SCQF 7
LLO8	Work effectively under guidance as part of a team in practical tasks.	SC5	SCQF 7, SCQF 7
LLO9	Demonstrate responsibility for personal learning and time management.	SC5	SCQF 7, SCQF 7
LLO10	Show an understanding of the main areas of renewable energy engineering and specialised knowledge in some areas of renewable energy, mechanical and electrical engineering.	SC1	SCQF 8
LLO11	Select suitable materials, equipment, engineering technologies and processes for engineering applications and projects.	SC1	SCQF 8
LLO12	Demonstrate ability in use of industry standard Engineering design and drawing software.	SC2	SCQF 8
LLO13	Use a range of laboratory skills, engineering practices and/or material to complete laboratory work	SC2	SCQF 8
LLO14	Carry out analysis of defined and/or routine problems and issues in renewable energy, mechanical and electrical systems	SC3	SCQF 8, SCQF 8
LLO15	Use scientific, mathematical, and engineering approaches to formulate and solve renewable energy, mechanical and electrical engineering problems.	SC3	SCQF 8
LLO16	Convey complex information relating to renewable energy, mechanical and electrical engineering in a clear, concise, and well-structured way using a range of forms of communication, including written and oral, and in both familiar and unfamiliar contexts.	SC4	SCQF 8
LLO17	Apply a range of numerical and graphical skills to process data obtained through theoretical study, laboratory experiments and engineering practices.	SC4	SCQF 8
LLO18	Select and choose ICT applications to process and obtain a variety of information and data in renewable, mechanical and electrical engineering context	SC4	SCQF 8
LLO19	Work effectively and under guidance, with fellow students and lab tutors, in laboratory and practical activities to acquire an understanding of current engineering practice.	SC5	SCQF 8
LLO20	Manage, under guidance, ethical and professional issues in accordance with current professional and/or ethical codes or practices.	SC5	SCQF 8
LLO21	Demonstrate a critical understanding of a range of concepts, principles, theories and terminology of renewable energy engineering.	SC1	SCQF 9
LLO22	Demonstrate knowledge of one or more specialised areas in renewable energy engineering which is informed by current development.	SC1	SCQF 9
LLO23	Demonstrate a critical understanding of project management, team-working, leadership, sustainability equality, diversity and inclusion in engineering practice.	SC1	SCQF 9
	Demonstrate principal professional skills, engineering practices and design skills in complex laboratory work and engineering projects with		

LLO24	comprehensive consideration of applicable health and safety, security, cultural, societal, and environmental matters.	SC2	SCQF 9
LLO25	Demonstrate the ability to assess ideas, concepts, information and issues in renewable energy, mechanical and electrical engineering.	SC3	SCQF 9
LLO26	Assess the performance and behaviour of techniques, approaches, technologies, and systems used in renewable energy, mechanical and electrical engineering based on information and data from laboratory and project work.	SC3	SCQF 9
LLO27	Draw on a range of sources and information while making judgements.	SC3	SCQF 9
LLO28	Present or communicate formally and informally, information on main topics in renewable energy, mechanical and electrical engineering to a range of audiences, including academic tutors, lab tutors, technicians, and fellow students.	SC4	SCQF 9
LLO29	Demonstrate academic communication skills to present work in a variety of forms (including coursework, project reports and portfolios) clearly, logically and in a well-structured manner.	SC4	SCQF 9
LLO30	Interpret, use and evaluate numerical and graphical data, obtained through theoretical study and practical work to achieve goals and targets.	SC4	SCQF 9
LLO31	Adopt a holistic and proportionate approach to the mitigation of security risks	SC5	SCQF 9
LLO32	Plan and record and reflect on self-learning and development as the foundation for lifelong learning/CPD	SC5	SCQF 9
LLO33	Function effectively as an individual, and as a member or leader of a team	SC5	SCQF 9
LLO34	Demonstrate initiative and independence in some activities at a professional level in renewable energy, electrical and mechanical engineering.	SC5	SCQF 9
LLO35	Work effectively under the guidance of specialists in laboratory exercises; group activities and projects; and engineering practices of renewable energy engineering.	SC5	SCQF 9
LLO36	Develop a critical understanding of a number of more specialised areas in Mechanical Engineering	SC1	SCQF 10
LLO37	Develop a detailed knowledge of one or more specialisms, some of which is at the forefront of the development of the industries related to mechanical engineering.	SC1	SCQF 10
LLO38	Discuss the role of quality management systems and continuous improvement in the context of complex problems	SC2	SCQF 10
LLO39	Apply an integrated or systems approach to the solution of complex problems in mechanical engineering	SC2	SCQF 10
LLO40	Use a risk management process to identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity in mechanical engineering	SC2	SCQF 10
LLO41	Use practical laboratory and workshop skills to investigate complex problems in mechanical engineering	SC2	SCQF 10
LLO42	Select and apply appropriate materials, equipment, engineering technologies and processes, recognising their limitations	SC2	SCQF 10
LLO43	Apply knowledge of engineering management principles, commercial context, project and change management, and relevant legal matters including intellectual property rights	SC3	SCQF 10
LLO44	Apply knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems in Mechanical Engineering. Some of the knowledge will be at the forefront of Mechanical Engineering	SC3	SCQF 10
LLO45	Analyse complex problems in mechanical engineering to reach substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles	SC3	SCQF 10
LLO46	Select and apply appropriate computational and analytical techniques to model complex problems in mechanical engineering, recognising the limitations of the techniques employed	SC3	SCQF 10
LLO47	Select and evaluate technical literature and other sources of information to address complex problems in mechanical engineering	SC3	SCQF 10

LLO48	Design solutions for complex problems in mechanical engineering that meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards	SC4	SCQF 10
LLO49	Communicate effectively on complex engineering matters with technical and non-technical audiences	SC5	SCQF 10
LLO50	Evaluate the environmental and societal impact of solutions to complex problems in mechanical engineering and minimise adverse impacts	SC5	SCQF 10
LLO51	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct	SC5	SCQF 10
LLO52	Adopt an inclusive approach to engineering practice and recognise the responsibilities, benefits and importance of supporting equality, diversity and inclusion	SC5	SCQF 10

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10

		SCQF 11 - SCQF 11
		SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering
Part Time On Campus	Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Stage 1 / Semester 1**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE1003	Create and Build	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	30	Level 7
Full Time On Campus	CE1005	Mathematical Thinking for Engineers	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 7
Full Time On Campus	CE1004	Statics and Structural Equilibrium	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 7

Stage 1 / Semester 2**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE1006	Design and Innovate	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	30	Level 7
Full Time				Hossein				

On Campus	CE1007	Engineering Analysis: Materials and Motion	Routes A, B, C	Zanganeh (hz9841)	Yes	1.0	15	Level 7
Full Time On Campus	CE1008	Foundations of Thermofluids	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 7

Stage 2 / Semester 1**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE2006	Design of Mechanical Systems	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8
Full Time On Campus	CE2007	Electromechanical Energy Conversion	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8
Full Time On Campus	CE2008	Mathematical Modelling and Analysis	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0469 - Electives 1	Mechanical Engineering - Electives Group 1	Routes A, B, C	Hossein Zanganeh (hz9841)		1.0	15	

Stage 2 / Semester 2**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE2013	Applied Thermofluids: Energy and Flow	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8
Full Time On Campus	CE2014	Engineering Analysis: Structures and Vibration	Routes A, B, C	Hossein Zanganeh	Yes	1.0	15	Level 8

Campus				(hz9841)				
Full Time On Campus	CE2012	Robot Wars	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0469 - Electives 2	Mechanical Engineering - Electives Group 2	Routes A, B, C	Hossein Zanganeh (hz9841)		1.0	15	

Stage 2 / Semester 3**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0469 - Electives 6	Mechanical Engineering - Electives Group 6	Routes A, B, C	Hossein Zanganeh (hz9841)		1.0	120	

Stage 3 / Semester 1**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE3024	Engineering Project Planning and Professional Practice	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 9
Full Time On Campus	CE3010	Engineers, Innovation and Society	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 9
Full Time On Campus	CE3021	Heat Transfer and Turbomachinery	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 9
Full Time On Campus	CM3151	Machine Learning	Route A	David Corsar (dc9396)	Yes	1.0	15	Level 9
Full Time On Campus	CE3013	Mechatronics	Route B	James Philp (jmntp5721)	Yes	1.0	15	Level 9
Full Time				Hossein				

On Campus	CE3018	Oil and Gas Drilling and Well Completion Engineering	Route C	Zanganeh (hz9841)	Yes	1.0	15	Level 9
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Stage 3 / Semester 2**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE3019	Circular Economy	Route C	Hossam Eldessouky	Yes	1.0	15	Level 9
Full Time On Campus	CE3012	Control Engineering	Routes A, B, C	James Philp (jmn5721)	Yes	1.0	15	Level 9
Full Time On Campus	CE3015	Engineering Group Design Project	Routes A, B, C	Hossein Zanganeh (hz9841)	No	1.0	15	Level 9
Full Time On Campus	CE3016	Integrating Renewable Energy: Smart Grids, Energy Storage and Hydrogen Systems	Route B	Naomi Turner (net5776)	Yes	1.0	15	Level 9
Full Time On Campus	CE3017	Renewable Energy Engineering	Route A	Naomi Turner (net5776)	Yes	1.0	15	Level 9
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0469 - Electives 3	Mechanical Engineering - Electives Group 3	Route A	Hossein Zanganeh (hz9841)		1.0	15	
Full Time On Campus	0469 - Electives 4	Mechanical Engineering - Electives Group 4	Route C	Hossein Zanganeh (hz9841)		1.0	15	
Full				Hossein				

Time On Campus	0469 - Electives 5	Mechanical Engineering - Electives Group 5	Route B	Zanganeh (hz9841)		1.0	15	
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Stage 3 / Semester 3**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Elective									
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>	
Full Time On Campus	0469 - Electives 6	Mechanical Engineering - Electives Group 6		Hossein Zanganeh (hz9841)		1.0	120		

Stage 4 / Semester 1**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE4009	Advanced Control and Instrumentation	Route B	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
Full Time On Campus	CE4003	Computational Mechanics	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
Full Time On Campus	CE4005	Electrical Power Machines and Applications	Route B	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
Full Time On Campus	CE4010	Failure Analysis	Route A	Hossam Eldessouky	Yes	1.0	15	Level 10
Full Time On Campus	CE4004	Offshore Hydrodynamics and Structural Loading	Route C	Naomi Turner (net5776)	Yes	1.0	15	Level 10
Full Time On Campus	CE4007	Petroleum Systems and Production Engineering	Route C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
Full Time On Campus	CE4011	Safety, Risk and Principles of Engineering Management	Route A	Hossam Eldessouky	Yes	1.0	15	Level 10
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full				Hossein				

Time On Campus	0469 - Electives 9	Mechanical Engineering - Elective Group 9	Route C	Zanganeh (hz9841)		1.0	15	
Full Time On Campus	0469 - Electives 10	Mechanical Engineering - Electives Group 10	Route B	Hossein Zanganeh (hz9841)		1.0	15	
Full Time On Campus	0469 - Electives 8	Mechanical Engineering - Electives Group 8	Route A	Hossein Zanganeh (hz9841)		1.0	15	

Stage 4 / Semester 2**Full Time On Campus**

Route A - Mechanical Engineering Route B - Mechanical and Electrical Engineering Route C - Mechanical and Offshore Engineering

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time On Campus	CE1003	Create and Build	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	30	Level 7
1	Semester 1	Core	Full Time On Campus	CE1005	Mathematical Thinking for Engineers	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CE1004	Statics and Structural Equilibrium	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CE1006	Design and Innovate	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	30	Level 7
1	Semester 2	Core	Full Time On Campus	CE1007	Engineering Analysis: Materials and Motion	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CE1008	Foundations of Thermofluids	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 7
2	Semester 1	Core	Full Time On Campus	CE2006	Design of Mechanical Systems	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CE2007	Electromechanical Energy Conversion	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CE2008	Mathematical Modelling and Analysis	Routes A, B, C	Hossein Zanganeh	Yes	1.0	15	Level 8

			Campus				(hz9841)					
2	Semester 1	Elective	Full Time On Campus	0469 - Electives 1	Mechanical Engineering - Electives Group 1	Routes A, B, C	Hossein Zanganeh (hz9841)		1.0	15		
2	Semester 2	Core	Full Time On Campus	CE2013	Applied Thermofluids: Energy and Flow	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8	
2	Semester 2	Core	Full Time On Campus	CE2014	Engineering Analysis: Structures and Vibration	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8	
2	Semester 2	Core	Full Time On Campus	CE2012	Robot Wars	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 8	
2	Semester 2	Elective	Full Time On Campus	0469 - Electives 2	Mechanical Engineering - Electives Group 2	Routes A, B, C	Hossein Zanganeh (hz9841)		1.0	15		
2	Semester 3	Elective	Full Time On Campus	0469 - Electives 6	Mechanical Engineering - Electives Group 6	Routes A, B, C	Hossein Zanganeh (hz9841)		1.0	120		
3	Semester 1	Core	Full Time On Campus	CE3024	Engineering Project Planning and Professional Practice	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 9	
3	Semester 1	Core	Full Time On Campus	CE3010	Engineers, Innovation and Society	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 9	
3	Semester 1	Core	Full Time On Campus	CE3021	Heat Transfer and Turbomachinery	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 9	
			Full									

3	Semester 1	Core	Time On Campus	CM3151	Machine Learning	Route A	David Corsar (dc9396)	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CE3013	Mechatronics	Route B	James Philp (jmdp5721)	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CE3018	Oil and Gas Drilling and Well Completion Engineering	Route C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CE3019	Circular Economy	Route C	Hossam Eldessouky	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CE3012	Control Engineering	Routes A, B, C	James Philp (jmdp5721)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CE3015	Engineering Group Design Project	Routes A, B, C	Hossein Zanganeh (hz9841)	No	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CE3016	Integrating Renewable Energy: Smart Grids, Energy Storage and Hydrogen Systems	Route B	Naomi Turner (net5776)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CE3017	Renewable Energy Engineering	Route A	Naomi Turner (net5776)	Yes	1.0	15	Level 9
3	Semester 2	Elective	Full Time On Campus	0469 - Electives 3	Mechanical Engineering - Electives Group 3	Route A	Hossein Zanganeh (hz9841)		1.0	15	
3	Semester 2	Elective	Full Time On	0469 - Electives 4	Mechanical Engineering - Electives Group 4	Route C	Hossein Zanganeh		1.0	15	

			Campus				(hz9841)				
3	Semester 2	Elective	Full Time On Campus	0469 - Electives 5	Mechanical Engineering - Electives Group 5	Route B	Hossein Zanganeh (hz9841)		1.0	15	
3	Semester 3	Elective	Full Time On Campus	0469 - Electives 6	Mechanical Engineering - Electives Group 6		Hossein Zanganeh (hz9841)		1.0	120	
4	Semester 1	Core	Full Time On Campus	CE4009	Advanced Control and Instrumentation	Route B	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CE4003	Computational Mechanics	Routes A, B, C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CE4005	Electrical Power Machines and Applications	Route B	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CE4010	Failure Analysis	Route A	Hossam Eldessouky	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CE4004	Offshore Hydrodynamics and Structural Loading	Route C	Naomi Turner (net5776)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CE4007	Petroleum Systems and Production Engineering	Route C	Hossein Zanganeh (hz9841)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CE4011	Safety, Risk and Principles of Engineering Management	Route A	Hossam Eldessouky	Yes	1.0	15	Level 10
			Full				Hossein				

4	Semester 1	Elective	Time On Campus	0469 - Electives 9	Mechanical Engineering - Elective Group 9	Route C	Zanganeh (hz9841)		1.0	15	
4	Semester 1	Elective	Full Time On Campus	0469 - Electives 10	Mechanical Engineering - Electives Group 10	Route B	Hossein Zanganeh (hz9841)		1.0	15	
4	Semester 1	Elective	Full Time On Campus	0469 - Electives 8	Mechanical Engineering - Electives Group 8	Route A	Hossein Zanganeh (hz9841)		1.0	15	
4	Semester 2	Core	Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	
Aberdeen, United Kingdom	On-Campus	Part Time	

Sources of Further Information

#	URL and Description
1	https://www.rgu.ac.uk/study/academic-schools/school-of-computing-engineering-and-technology School of Computing, Engineering and Technology